

Proposal: Replace the sourcing reverse block scan with an index lookup

Status	Proposal
Area	JetStream stream sourcing (<code>server/stream.go</code>) + store index (<code>server/filestore.go</code> , <code>server/memstore.go</code>)
Date	2026-06-06
Companion	<code>sourcing-durable-resume.md</code> (lifecycle, durable consumer, triggers)
Target	<code>startingSequenceForSources</code> (<code>stream.go:4694</code>) and <code>setStartingSequenceForSources</code> (<code>stream.go:4566</code>)

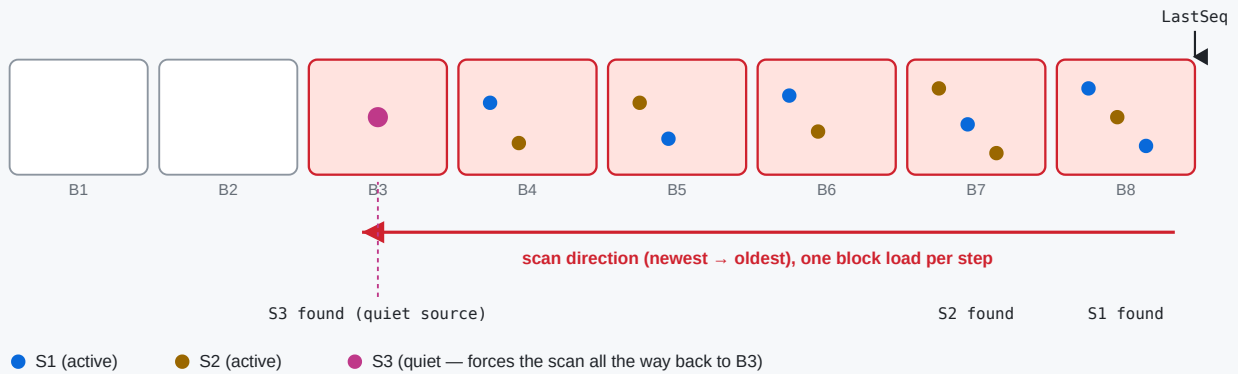
1. The problem in one paragraph

On hub-stream **leader election / restart**, `setupSourceConsumers` calls `startingSequenceForSources` **unconditionally** (`stream.go:4821`). That function reconstructs each source's last sourced sequence by walking the stream's **own store backwards from `LastSeq`**, one block at a time (`LoadPrevMsgMulti` , `filestore.go:9425`), reading the `Nats-Stream-Source` header off matching messages, until **every** source has been located.

Note: the walk is not as naive as message-by-message — within each block the per-block index (`fss`) skips non-matching messages, and the per-source sublist narrows as sources are found. But `LoadPrevMsgMulti` still **visits every block** from `LastSeq` back to the match (loading/decompressing cold ones), and the outer loop **re-walks** as the sublist narrows. So the cost grows with both the number of blocks back to the least-recently-active source (worst case the **entire store**) and the number of sources — all under `mset.mu` .

The reverse block scan: rebuild each source's last sequence from stored headers

startingSequenceForSources walks back from LastSeq, LoadPrevMsgMulti loads (+decompresses) a block at a time until every source is found.



Why it can be slow on a limits-based stream

A single quiet/sparse source (S3) forces the scan past every newer block — worst case the entire stream. Each uncached block is read from disk, decrypted and decompressed. With many sources and a large store, leader election stalls under mset.mu while this runs.

2. The key insight

The store **already maintains a per-subject index** that can return the last sequence for a subject *without* walking messages:

- `fs.psim` — per-subject info, including `lblk` = the **last block index** that contains each subject (`filestore.go:3843-3852` , `8980-8984`).
- `mb.fss` — per-block subject state with `Last` per subject (`filestore.go:9003-9015`).

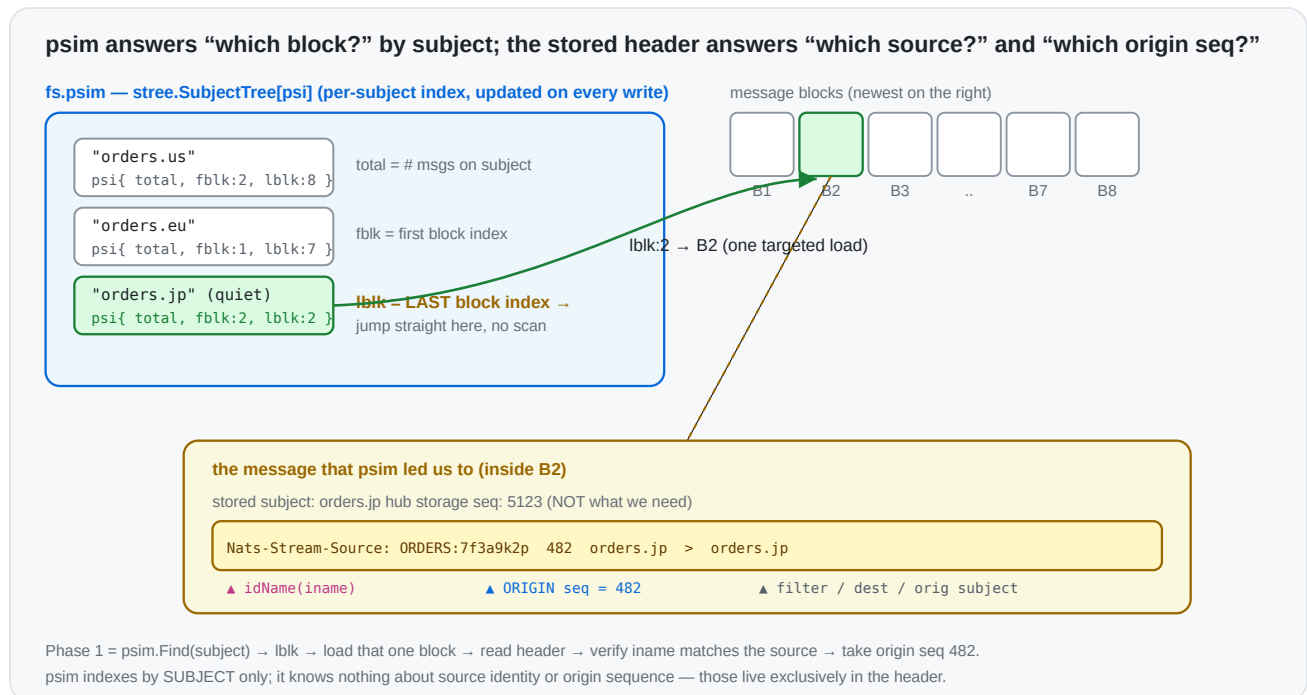
This is what powers `LoadLastMsg(subject) → loadLast ( filestore.go:8939 )`: for a literal subject it jumps straight to `info.lblk` and reads that subject's last message — **one targeted block load**, regardless of how far back it is. `MultiLastSeqs ( filestore.go:3806 )` does the same for a batch of filters in a single index pass.

The current scan ignores this index and instead does a generic backward message walk. The header it needs (the **origin** sequence) is right there in the message that `LoadLastMsg` already returns.

Why a plain "last seq for subject" isn't enough on its own: `si.sseq` must be the **origin** stream sequence taken from the `Nats-Stream-Source` header, not the hub's storage sequence. So we use the index to *find* the source's last stored message in $O(1)$, then read the origin seq from its header.

3. Background: the subject index (`psim` + `fss`) and the source header

The filestore keeps a **two-level subject index**. Phase 1 stands entirely on it, so it's worth being precise about what each level is.



3.1 `psim` — the store-wide "which blocks hold this subject" map

`fs.psim` (`filestore.go:195`) is a `stree.SubjectTree[psi]` — a subject-token tree keyed by the full message subject — whose value is a tiny per-subject record (`filestore.go:168`):

```
type psi struct {
    total uint64 // how many messages exist for this subject across the whole s
    fblk  uint32 // index of the FIRST block that still contains this subject
    lblk  uint32 // index of the LAST block that still contains this subject
}
```

It is maintained incrementally on every write (`filestore.go:4933-4943`): on store, the subject's entry is created (`total:1, fblk=lblk=current block`) or its `total` is bumped and `lblk` advanced to the current block; on removal/expiry the counts and `fblk` / `lblk` are walked forward as blocks empty out. Because `psim` is a *subject tree*, it supports both exact `Find(subject)` and wildcard `Match(filter)` in time proportional to the subject token structure — **not** to the number of messages.

The single field that makes Phase 1 cheap is `lblk` : given a source's subject, `psim.Find` returns the **exact last block** that contains it. There is no need to scan newer blocks — most of which don't contain that subject at all. (`MultiLastSeqs` , `filestore.go:3806` , uses the same `psim` to answer a *batch* of filters in one index pass.)

3.2 `fss` — the per-block "first/last seq of this subject in this block" map

`psim` gets us to a block; `mb.fss` (`filestore.go:239`) finishes the job *inside* that block. Each `msgBlock` carries its own `stree.SubjectTree[SimpleState]` mapping subject → (`store.go:180`):

```
type SimpleState struct {
    Msgs  uint64 // messages for this subject in THIS block
    First uint64 // first sequence for this subject in this block
    Last  uint64 // last  sequence for this subject in this block
    // (firstNeedsUpdate / lastNeedsUpdate: lazily recomputed when interior dele
}
```

So the two levels compose into an O(1)-ish lookup, which is exactly what `loadLast` (`filestore.go:8980-9034`) does:

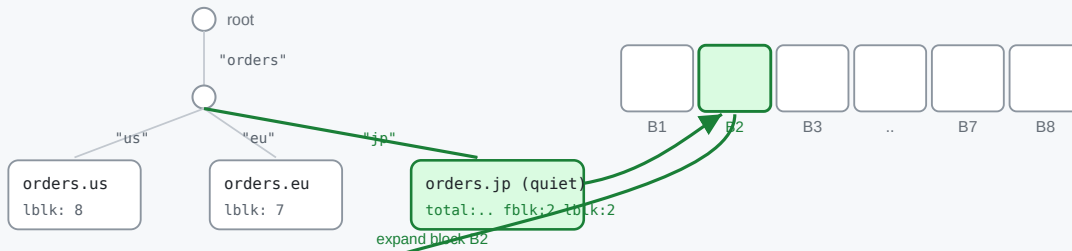
```
psim.Find(subject) → info.lblk → block → block.fss.Find(subject) → ss.Last
```

`fss` is what gives us the precise sequence within the block (and it correctly skips interior-deleted messages via `lastNeedsUpdate` / `recalculateForSubj`). On a cold block, `ensurePerSubjectInfoLoaded` loads/derives `fss` for that **one** block — the cost we pay once per resolved source, versus the current scan paying it for every block back to the oldest source.

Two-level traversal: psim subject tree → the block → that block's fss subject tree → the message

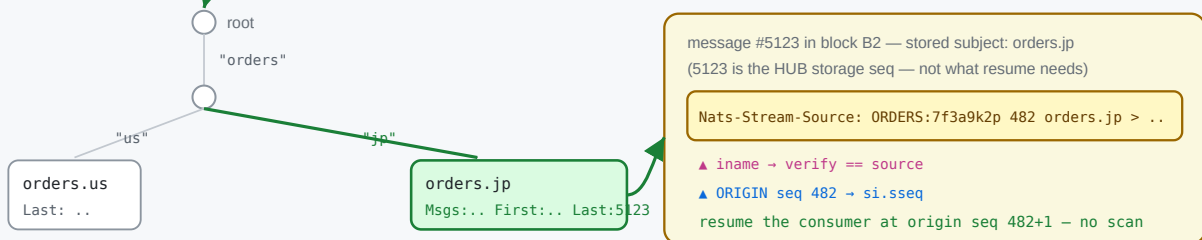
1 fs.psim — store-wide subject tree (subject → psi{total, fblk, lblk})

2 follow lblk → the last block holding that subject



3 B2.fss — per-block subject tree (subject → SimpleState{Msgs, First, Last})

4 load msg at Last (hub seq 5123) → read its header



3.3 Why each message carries a `Nats-Stream-Source` header

psim / fss index by **subject**. They know nothing about *which* source produced a stored message or *where in the origin stream* it came from — and those are precisely the two facts resume needs. That information is carried only in the per-message header `Nats-Stream-Source` (`JSStreamSource`, `stream.go:635`), written by `genSourceHeader` (`stream.go:4470`) as:

```
Nats-Stream-Source: <idName> <originSeq> <filter> <destination> <origSubject>
                    |           |
                    |           └ the ORIGIN stream's sequence – what we must
                    └ origin stream name (+ domain/consumer hash); with filter
                        reconstructs the source's full iname (parsed by stream)
```

This header is needed for three independent reasons:

1. **Source attribution.** A hub message is stored under its *destination* subject; the storage layer records nothing about its origin. When two sources can land on the same/overlapping subject, only the header's `iname` says which source a given message belongs to — hence Phase 1's `iname == si.iname` check.
2. **Origin-sequence tracking.** The sourcing consumer is (re)created on the **origin** with `DeliverByStartSequence`, which lives in the *origin*'s sequence space. The hub's own storage sequence (e.g. 5123) is unrelated to the origin sequence (e.g. 482). Only the

header preserves the origin sequence, so it is the *only* place a resume point can be recovered from without contacting the origin.

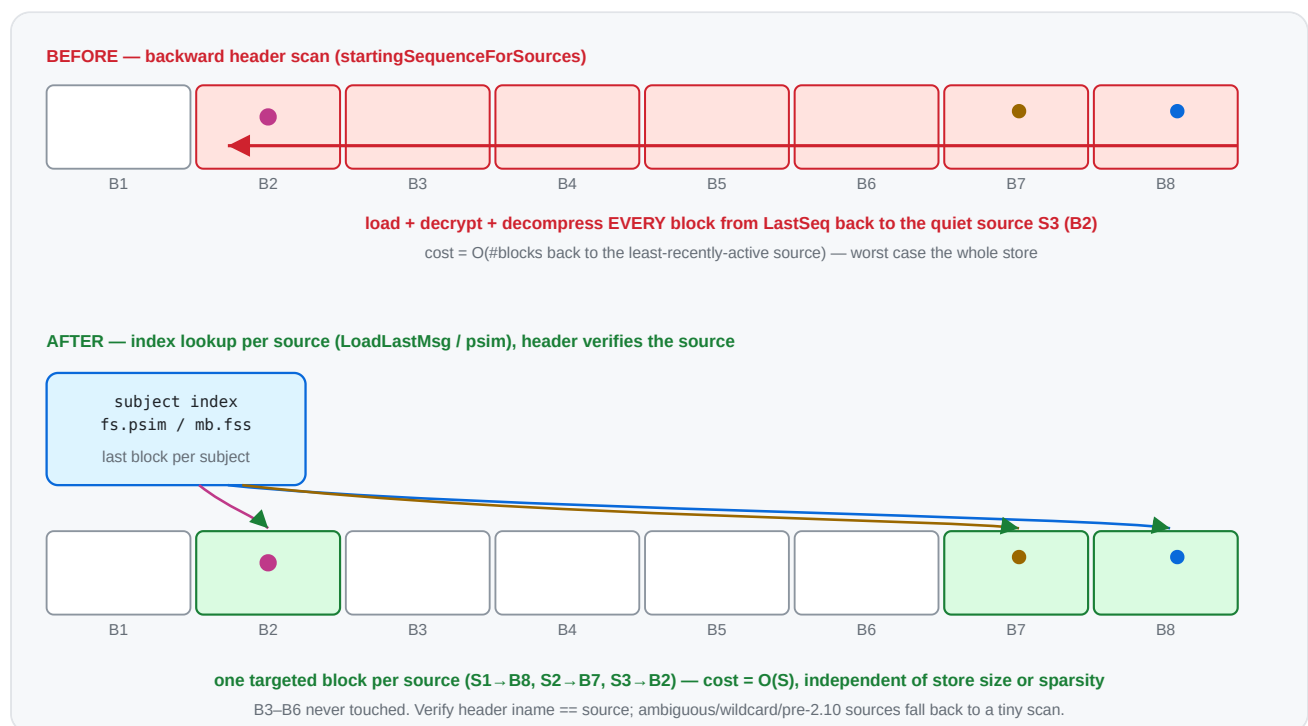
3. Loop / daisy-chain handling & dedup. When re-sourcing ($A \rightarrow B \rightarrow C$),

`processInboundSourceMsg` strips the inbound `JSSource` and stamps its own (`stream.go:4398`, `4405`), so each hop's provenance is well-defined and re-sourced duplicates can be recognised.

This is the crux of the whole proposal: **the index tells us *which block* in $O(1)$; the header (read from that one message) tells us *which source* and *which origin seq***. Today's scan reads the header off *every* message on the way back because it never consults `psim / fss` to jump directly to the right one.

4. Proposed algorithm

Two phases. Phase 1 resolves the common case with index lookups; Phase 2 keeps today's scan only for sources that genuinely can't be attributed by subject.



For each source, its **stored subject** is the `FilterSubject` for a plain source, or — for a source with a **single, concrete (non-templated)** subject transform — the transform **Destination**.

Templated transform destinations contain `{{...}}` mapping tokens (the rendered subject varies per message), and partial wildcards in a destination are rejected at config time, so a destination is concrete unless it is empty, `>`, or contains `{{`.

Phase 1 – index lookup (per source, $O(1)$ targeted load):
for each source `si`:

```

    subj := si.FilterSubject // plain source
    if si has exactly one transform: subj := transform.Destination
    if subj == "" || subj == ">" || subj contains "{": defer to Phase 2; continue
    sm := store.LoadLastMsg(subj) // jumps via psim, ~1 block
    if sm has a JSStreamSource header:
        (_, iname, sseq) := streamAndSeq(header)
        if iname == si.iname: // self-verifying attribution
            si.sseq = sseq; mark resolved
    // anything not resolved (no header / header for a different source / wildcard
    // multi-or-templated transform / pre-2.10) → Phase 2

```

Phase 2 – fallback reverse scan (only for the unresolved subset):

run today's LoadPrevMsgMulti loop, but with the sublist built from ONLY the unresolved sources, so it narrows immediately and stops as soon as they are found

The header check makes Phase 1 **self-correcting**: if a subject is shared between sources, or overlaps a direct publish on the hub, or the last message has no source header, the lookup simply doesn't match and that source drops to Phase 2 — never producing a wrong answer.

5. Before → after (code)

Before (`startingSequenceForSources` , condensed — `stream.go:4694`)

```

mset.resetSourceInfo()
// build a sublist of ALL sources' subjects
refreshSublist() // every source
for last := state.LastSeq; ; { // walk the WHOLE store backwards
    sm, seq, err := mset.store.LoadPrevMsgMulti(sl, last, &smv) // loads/decompresses
    if err == ErrStoreEOF || err != nil { break }
    last = seq - 1
    ...
    _, iName, sseq := streamAndSeq(bytesToString(ss))
    update(iName, sseq) // narrows sublist as sources are resolved
    if len(seqs) == expected { return } // stop only when ALL sources resolved
}

```

Cost: O(blocks back to the least-recently-active source) block loads → worst case the entire store.

After

```
mset.resetSourceInfo()
var smv StoreMsg
unresolved := map[string]*sourceInfo{}

// Phase 1: targeted index lookup per source.
for _, ssi := range mset.cfg.Sources {
    si := mset.sources[ssi.iname]
    if si == nil { continue }
    var subj string // FilterSubject, or single c
    switch {
    case len(ssi.SubjectTransforms) == 0: subj = ssi.FilterSubject
    case len(ssi.SubjectTransforms) == 1: subj = ssi.SubjectTransforms[0].Destin
    }
    if subj == _EMPTY_ || subj == fwcs || strings.Contains(subj, "{{") {
        unresolved[ssi.iname] = si // can't attribute by a single
        continue
    }
    sm, err := mset.store.LoadLastMsg(subj, &smv) // 0(1) via psim/fss, returns
    if err != nil || sm == nil || len(sm.hdr) == 0 {
        unresolved[ssi.iname] = si
        continue
    }
    if ss := sliceHeader(JSStreamSource, sm.hdr); len(ss) > 0 {
        if _, iName, sseq := streamAndSeq(bytesToString(ss)); iName == ssi.iname
            si.sseq, si.dseq = sseq, 0 // verified – resolved without
            continue
        }
    }
    unresolved[ssi.iname] = si
}

// Phase 2: only the leftovers pay the (now tiny) reverse scan.
if len(unresolved) > 0 {
    mset.reverseScanForSources(unresolved) // = today's loop, scoped to
}
```

Cost: Phase 1 = $O(S)$ targeted lookups (recent sources share/cache the last block; a quiet source is **one** targeted load, not a walk). Phase 2 only runs for ambiguous sources, with a smaller sublist.

Multi-destination transform sources: call `LoadLastMsg` for each destination and keep the highest origin `sseq`, or defer them to Phase 2. Either is fine; deferring is simplest to start.

6. Before → after (complexity)

	Before	After
Index used	none (generic message walk)	<code>psim</code> / <code>fss</code> per-subject last-block index
Blocks loaded	back to least-active source ($\leq$ all blocks)	~1 per source, only the blocks actually holding a source's last msg
Sensitivity to a quiet source	drags the scan back across the whole store	none — jumps straight to its block
Sensitivity to store size	linear in blocks scanned	none
Worst case still possible?	normal case	only if every source is <code>> /wildcard/shared-subject/pre-2.10</code> (→ Phase 2 == today)
Held lock	<code>mset.mu</code> for the whole walk	<code>mset.mu</code> for O(S) lookups; Phase 2 only for leftovers

The edge → hub topology (each edge mapped to a distinct subject/domain) lands entirely in Phase 1 → the backward scan disappears for that case.

7. Phase 2: keeping the fallback scan narrow

Phase 1 removes most sources from the picture, but whatever remains still runs the reverse `LoadPrevMsgMulti` scan. That scan's whole speed advantage depends on the **sublist being narrow** — `prevMatchingMulti` uses each block's `fss` to skip blocks that contain none of the sublist's subjects. A single `>` (full wildcard) in the sublist defeats that: it matches every subject, so no block can be skipped and the scan visits every block back to the match.

Phase 2 sublist narrowing: one transform source no longer poisons the whole scan

BEFORE — transform/catch-all source injects ">" (empty FilterSubject)



AFTER — phase 1 resolves wildcard transforms; phase 2 sublist uses destinations

step 1: phase 1 collapses wildcard(/)\$N tokens and resolves via the index

tout.{{wildcard(1)}} → tout.* → LoadLastMsg("tout.*") ✓

step 2: only genuinely-unresolved sources remain; sublist stays narrow



Remaining ">" cases (by design): a genuine catch-all source (empty filter, no transform) or an exotic transform (partition/split).

What used to inject >

1. **Every transform source.** A transform source has an empty `FilterSubject` (mutually exclusive with `SubjectTransforms`, `stream.go:1976`), so the old `refreshSublist` did `sl.Insert(fwcs)`. One transform source in the unresolved set widened the whole sublist to `>` → full backward scan.
2. **Templated transforms were deferred to Phase 2 at all.** Their *rendered* destination is a narrow wildcard (e.g. `tout.{{wildcard(1)}}` always lands on `tout.*`), so they don't need a full scan.
3. **Dead entries.** Phase 2 also inserted `si.sfs` — the transform *source* filters (e.g. `tin.*`) — which never match stored subjects (messages are stored under the *destination*).

The two changes

- **Resolve wildcard transforms in Phase 1.** `transformUntokenize(dest)` collapses `wildcard()` / `$N` mapping tokens to a subject wildcard (`tout.{{wildcard(1)}} → tout.*`); `LoadLastMsg` accepts that wildcard and the header- `iname` check still guards attribution. So the common templated transforms resolve in Phase 1 and never reach the scan.
- **Build the Phase 2 sublist from destinations.** For any transform source that *does* reach Phase 2 (e.g. its wildcard space is shared with another source, so Phase 1 saw a header for the other one), insert the destination's wildcard form instead of `>`, and drop the dead `si.sfs` inserts.

What still forces > (by design)

Only a source that *genuinely* needs a broad match:

- a **catch-all source** — empty `FilterSubject`, no transform — sourcing every subject of its origin; and
- an **exotic transform** (`partition` / `split` / `slice` /...) whose rendered destination isn't a subject wildcard, so `transformUntokenize` can't reduce it (it leaves the `{{...}}` token in place, which we detect and fall back to `>`).

Both are uncommon for the edge → hub fan-in. Distinct subjects, wildcard filters, and `wildcard()` / `$N` transforms all either resolve in Phase 1 or keep the Phase 2 sublist narrow enough to skip blocks.

Correctness note: this is purely about *narrowing*, never about *missing* a source. If a narrowed subject is wrong (shared space), the header check fails and the source stays in the unresolved set; an exotic/empty case falls back to `>` . So Phase 2 still finds every source it did before — it just loads far fewer blocks getting there.

8. Correctness & edge cases

- **Same semantics.** Today's scan records, per source, the **most recent** stored message's origin seq (first hit scanning backward). `LoadLastMsg` returns exactly that message. Deleted/interior messages are handled by `loadLast` (it skips `dmap` entries and walks to the previous block if needed), matching the scan's behaviour.
- **Shared subjects / direct publishes.** Resolved safely by the header-iname verification → fall to Phase 2.
- **Wildcard / empty (`>`) filters.** Can't be attributed to one source by subject → Phase 2 (same as today for them).
- **Pre-2.10 headers** (no `iname`, only stream name): verification won't match → Phase 2, which keeps the existing stream-name matching path.
- **memstore.** `MemStore` has the simpler linear `LoadPrevMsgMulti`; `LoadLastMsg` there is also index-light, but memstore streams are small, so Phase 2 cost is negligible. (We can add a memstore `loadLast` fast path later if needed.)
- **setStartingSequenceForSources** (the `STREAM.UPDATE` path, `stream.go:4566`) gets the same Phase 1 treatment for the subset of sources it processes.

9. Risks

- **Index freshness.** `psim` / `fss` may lazily need a recalculation (`lastNeedsUpdate`, `recalculateForSubj`); `loadLast` / `MultiLastSeqs` already handle that, so we inherit

correct behaviour.

- **Behavioural parity.** Phase 1 must produce identical `si.sseq` to the old scan for non-ambiguous sources; this is the core test (below). Keeping Phase 2 byte-for-byte as today bounds the blast radius.
- **Encryption/compression.** Phase 1 still loads the one block holding a source's last message (decrypt/decompress), but only that block — no change in correctness, large reduction in volume.

10. Testing & validation

Prototype status — implemented

A working prototype of the two-phase resolver is implemented in `startingSequenceForSources` (`server/stream.go`), with:

- `TestJetStreamStartingSequenceForSourcesIndexFastPath` (`server/jetstream_sourcing_resume_test.go`) — three distinct-subject sources (index fast path) plus one **subject-transform** source (phase 2 fallback), each with a different origin depth and buried under 20k direct publishes. Asserts every recovered `si.sseq` equals the expected last origin sequence. **Passes.**
- `TestJetStreamStartingSequenceForSourcesAmbiguity` — a catch-all (empty filter) source and **two sources transformed onto the same destination subject** (shared stored subject), mixed with a distinct-subject source. Asserts the phase 2 fallback disambiguates all of them. **Passes.**
- `TestJetStreamSetStartingSequenceForSourcesIndex` — the same fast path applied to the `STREAM.UPDATE` twin `setStartingSequenceForSources` (distinct subject + catch-all + a **concrete-destination transform** source), clearing and recovering `sseq`. **Passes.**
- **Transform fast path:** a source with a single subject transform is resolved in Phase 1 by collapsing `wildcard()` / `$N` mapping tokens to a wildcard form (`transformUntokenize` , e.g. `tout.{{wildcard(1)}}` → `tout.*`) and doing `LoadLastMsg(wildcardForm)` , verified by the header. Only multi-transform and **exotic** transforms (`partition` / `split` / ... that don't reduce to a subject wildcard) still defer to Phase 2.
- **Phase 2 no longer degrades to `>` for transform sources.** Previously a transform source (empty `FilterSubject`) inserted a `>` catch-all into the sublist, defeating per-block skipping for the whole scan. The Phase 2 sublist is now built from the transform **destinations** (wildcard form), with the dead `si.sfs` source-filter inserts removed; only a genuine catch-all source or an exotic transform still forces `>` .
- The change was also applied to `setStartingSequenceForSources` itself (`stream.go:4566`), so both the leader-election and the config-update resume paths use the index fast path and the narrowed sublist.

- Tests (all pass, incl. `-race`): `...IndexFastPath`, `...Ambiguity` (catch-all + shared concrete destination), `...TemplatedTransform` (wildcard transform), `...SharedWildcardTransform` (two templated transforms sharing a `gout.*` space), `TestJetStreamSetStartingSequenceForSourcesIndex` (twin), and `TestJetStreamSourcingResumeAfterRolloutRestart` (end-to-end hard-restart, exactly-once resume).
- `BenchmarkJetStreamScanForSources` (existing, single source), `BenchmarkJetStreamScanForSourcesMulti` (16 sources spread across the store), `BenchmarkJetStreamSourceResumeLeafnodeFanIn` (8–512 edges feeding a hub, quiet edges buried under an all-sourced tail), `...FanInTailDepth` (fixed edges, varying tail depth — isolates the store-depth axis), `...PartitionFallback` (the by-design `>` path), and `...DeepStore` (direct store seeding up to 16/64/256 sources × 10M, for at-scale and cross-design comparison). See the tables below.
- Existing sourcing suite (`SourceBasics`, `SourceRemovalAndReAdd`, `WorkQueueSourceRestart`, `SourceWorkingQueueWithLimit`, `StreamSourceWithoutDuplicateWindow`, `MirrorAndSourcesFilteredConsumers`) still passes.

Measured (filestore, single-server)

Benchmark	Before (reverse scan)	After (phase 1)	Speedup
<code>ScanForSources</code> — 1 source, ~100k buried msgs	~8.5 µs/op	~5.5 µs/op	~1.5×
<code>ScanForSourcesMulti</code> — 16 sources, spread	~110.8 µs/op	~14.0 µs/op	~7.9×

The single-source gap is modest because `LoadPrevMsgMulti` already skips non-matching messages per block; the win scales with the number of sources (each avoids a re-walk via a direct `psim` jump), which is exactly the edge → hub fan-in case.

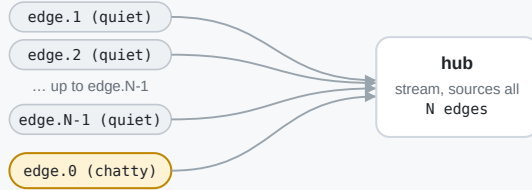
Leafnode fan-in resume (the leader-election case)

`BenchmarkJetStreamSourceResumeLeafnodeFanIn` models the scenario directly: a hub sources from N edge/leafnode streams, then one chatty edge sources a deep 100k tail that buries the other edges' last messages. Crucially the hub store is **entirely sourced edge subjects** — as a real fan-in is — so every block matches the recovery sublist and *none* can be skipped. This is the work a freshly elected leader runs (`startingSequenceForSources`) before it can resume sourcing. One op == one full recovery.

BenchmarkJetStreamSourceResumeLeafnodeFanIn — recovery on leader election

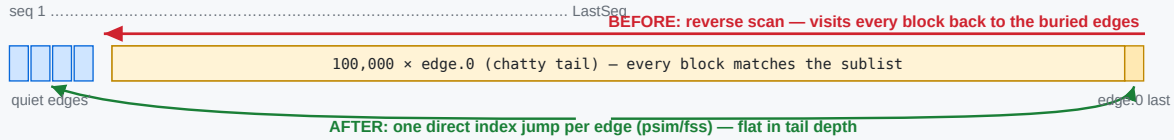
N edge/leafnode streams source into one hub; one chatty edge buries the quiet edges. Measures startingSequenceForSources (one op = one full resume).

1 • Setup — fan-in into the hub



Each quiet edge sends 1 message up front.
The chatty edge then sources a deep tail
tailDepth = 100,000
The hub store is entirely sourced edge
subjects — a real fan-in — so no block in
the tail can be skipped by per-block fss.

2 • Hub store after setup, and how each strategy traverses it



3 • What recovery does

BEFORE — reverse block scan

LoadPrevMsgMulti walks back from LastSeq. The tail is all in-sublist, so nothing is skipped → walks all 100k to reach the buried edges; rebuilds the sublist per source found.
cost ≈ tail depth + O(sources²) sublist rebuild

AFTER — per-subject index (phase 1)

LoadLastMsg(edge.i) jumps straight to each edge's last message via the psim/fss index; header verifies the source. No reverse scan, no sublist built.
cost ≈ N index lookups (flat in tail depth)

4 • Measured (filestore, single-server, benchstat n=6, p=0.002 — geomean -96.1%)

edges	before	after	Δ
8	7.27 ms	0.194 ms	-97.3%
32	7.96 ms	0.286 ms	-96.4%
128	13.2 ms	0.609 ms	-95.4%
512	71.1 ms	3.68 ms	-94.8%

before = store-bound (walks the tail);
after = source-bound (N lookups).
512: O(sources²) rebuild dominates before.

edges	Before (reverse scan)	After (phase 1)	Δ
8	7.27 ms ±96%	0.194 ms ±23%	-97.3%
32	7.96 ms ±6%	0.286 ms ±8%	-96.4%
128	13.2 ms ±6%	0.609 ms ±19%	-95.4%
512	71.1 ms ±22%	3.68 ms ±45%	-94.8%
geomean	15.3 ms	0.594 ms	-96.1%

(filestore, single-server, -benchtime=50x -count=6, benchstat; all rows p=0.002 (n=6). Both columns on the same shared host; "before" measured by splicing in the pre-optimization function from the parent commit. The host is noisy — note the ±96% spread on the edges=8 "before" outlier — but the per-row deltas and the -96.1% geomean are stable across runs.)

Two effects compound, and the table separates them:

- **Store-bound vs source-bound.** "Before" carries a ~10 ms floor independent of edge count — it must reverse-scan the whole 100k tail to reach the buried quiet edges. "After" is flat in tail depth and scales only with edge count (one LoadLastMsg index jump each).
- **The O(sources²) sublist rebuild.** The old Phase 2 rebuilt the whole sublist on every source it resolved (refreshSublist per update). At 512 sources that is ~260k inserts per recovery — the "before" jumps to ~70–100 ms and (in a separate -benchmem run)

49.8 MB / 674k allocs/op. Phase 1 resolves these without ever building a sublist, so allocations drop to **2,572/op** — ~260× fewer at 512 edges.

Caveat: this is a micro-benchmark of the recovery function, not a full Raft election — an actual election adds a fixed quorum/round-trip cost on top of *both* columns equally. An earlier draft buried the edges under *direct* (non-sourced) hub traffic; there the old per-block `fss` skipping rescued the scan (only ~5×), which is not representative of a pure fan-in. The committed benchmark uses an all-sourced tail so nothing is skippable.

Isolating the store-depth axis

`BenchmarkJetStreamSourceResumeFanInTailDepth` fixes the edge count (32) and varies only the buried tail depth. It makes the store-bound vs source-bound split explicit: the old scan grows linearly with depth while Phase 1 stays flat, so the speedup *grows* with how deep the store is.

tail depth	Before (reverse scan)	After (phase 1)	Speedup
50k	0.89 ms	0.30 ms	3.0×
100k	11.6 ms	0.97 ms	11.8×
200k	23.1 ms	1.13 ms	20.5×
400k	46.0 ms	1.11 ms	41.6×

"Before" doubles with the tail (11.6 → 23.1 → 46.0 ms) — textbook $O(\text{depth})$. "After" is flat at ~1 ms and its allocations stay at **~223/op** regardless of depth, versus ~3,400/op for the old scan. (The 50k "before" is sub-linear because at that size the buried edges still sit within the first couple of blocks the scan reaches; from 100k on, the linear walk dominates.)

At scale (10M), and vs. a persisted resume map

A parallel design (see §11) persists a `map[source]→lastSeq` and *reads* it on recovery instead of recomputing from the store. To compare on the same axis,

`BenchmarkJetStreamSourceResumeDeepStore` seeds the hub store directly — real end-to-end sourcing can't build a 10M-message store — with one `JStreamSource`-headered anchor per source plus a deep tail, and a built-in check verifies phase 1 resolves every source before timing. This pushes the reverse-scan baseline to 10M:

sources	tail	Before (reverse scan)	After (index recompute)	Speedup
16	100k	8.6 ms	0.265 ms	33×

sources	tail	Before (reverse scan)	After (index recompute)	Speedup
16	1M	96.9 ms	0.249 ms	389×
16	10M	972 ms	0.265 ms	3,668×
64	10M	974 ms	1.05 ms	928×
256	10M	991 ms	3.08 ms	322×

Both axes confirm the model: the reverse scan is $O(\text{depth})$ ($8.6 \rightarrow 97 \rightarrow 972$ ms across $100k \rightarrow 1M \rightarrow 10M$) and $\sim$ flat in source count (it walks the whole tail regardless — $972/974/991$ ms for $16/64/256$ at $10M$), while the index recompute is flat in depth (~ 0.27 ms at 16 sources from $100k$ to $10M$) and scales with source count (~ 12 μ s/source).

Cross-checking the persisted-map design's published numbers (different host, larger messages):

sources / tail	their scan baseline	their persisted map	our scan baseline	our index recompute
16 / 10M	5.03 s	2.08 μ s	0.972 s	0.265 ms

The two scan baselines have the same $O(\text{depth})$ shape; the $\sim 5\times$ absolute gap is environment (message size / host). The persisted map is $\sim 100\times$ faster than our index recompute *in absolute terms* because it never touches the store — a pure in-memory read — whereas the index still loads one block per source. Those two factors compose to explain why their headline multiplier ($\sim 2.4M\times$) dwarfs ours ($\sim 3,700\times$): $2.42M / 3,668 \approx 660 \approx 5.2$ (baseline gap) $\times 127$ (recompute vs map-read). The designs are **complementary, not competing** — see §11.

What still forces a full `>` scan in Phase 2 (by design)

After the above, Phase 2 only widens to `>` (no block skipping) when a source *genuinely* needs it:

- a **catch-all source** — empty `FilterSubject`, no transform — that sources every subject of its origin (it really can land anywhere, so a broad match is correct); and
- an **exotic transform** (`partition` / `split` / `slice` / ...) whose rendered destination is not a subject wildcard, so `transformUntokenize` can't reduce it to a matchable pattern.

Both are uncommon for the edge $\rightarrow$ hub fan-in. Everything else — distinct subjects, wildcard filters, `wildcard()` / `$N` transforms — either resolves in Phase 1 or narrows the Phase 2 sublist to a concrete or wildcard subject.

`BenchmarkJetStreamSourceResumePartitionFallback` measures this residual path with a `partition()` transform source (stored subjects `p.<n>`, which can't be reduced to a wildcard)

alongside several distinct edges (always resolved in Phase 1), over a deep 100k all-sourced store:

placement of the <code>></code> source	recovery	note
active (it's the chatty aggregator, last msg at LastSeq)	0.65 ms	the common case — the <code>></code> scan resolves in O(1) despite the fallback
buried (quiet early, a distinct edge forms the tail)	56.2 ms	residual worst case — the <code>></code> scan must walk the whole tail

The takeaway: the by-design `>` fallback is cheap whenever the catch-all/partition source is *active* (which is the normal state for an aggregator). It only degrades to a full scan when such a source is both present **and** has gone quiet under a deep tail — genuinely unavoidable, since a catch-all source has no single subject to index by. The distinct edges in the same stream are unaffected either way (Phase 1).

Testing plan & coverage

The recovery path has two layers worth testing separately: the **resolver** (`startingSequenceForSources` producing the right `si.sseq`) and the **end-to-end behaviour** (a real restart/election resumes sourcing exactly-once). Below is the current coverage and the prioritized gaps.

Layer 1 — resolver correctness (unit, calls the resolver directly).  in place: `...`

`IndexFastPath` (phase 1 + transform phase 2), `...Ambiguity` (catch-all + shared concrete destination), `...SetStartingSequenceForSourcesIndex` (update-path twin), `...`
`TemplatedTransform` , `...SharedWildcardTransform` .

Layer 2 — end-to-end recovery. Newly added:

`TestJetStreamSourcingResumeAfterRolloutRestart` — sources across both phases, hard-restarts the server (rolling-upgrade style), publishes a second batch, and asserts each origin's sourced sequences are exactly `1..N` (no gap = no missed resume; no duplicate = no re-sourced run). Passes under `-race` .

Prioritized gaps:

Pri	Area	Proposed test(s)	Guards against
P0	Clustered resume (R3)	<code>...ResumeAfterLeaderStepDown</code> — <code>mset.raftNode().StepDown()</code> , wait for new leader, publish more, assert exactly-once	resolver runs per-node on every election; the common production trigger, untested today

Pri	Area	Proposed test(s)	Guards against
P0	Property / differential	fuzz random source layouts (subjects, transforms, counts, interleaving, deletes) and assert the index resolver's <code>si.sseq ==</code> a brute-force reference linear scan	cheap broad coverage; catches resolver regressions the hand-written cases miss
P1	Store-state edges (seeded)	seed the store directly (as <code>...DeepStore</code> does) to build: pre-2.10 headers (empty <code>iname</code> , stream-name match), direct-publish/source subject overlap , and interior deletes/purge before a source's last message	the header- <code>iname</code> disambiguation and <code>loadLast</code> 's <code>dmap /prev-block</code> walk — hard to build via the JS client
P1	<code>memstore</code> path	run the resolver matrix on a <code>MemStore</code> -backed stream	the linear <code>LoadPrevMsgMulti</code> / index-light <code>LoadLastMsg</code> path is untested for sources
P1	Exotic transforms	<code>partition()</code> / <code>split()</code> source resume correctness (the by-design <code>></code> fallback)	the residual Tier 2 path; perf already benched (<code>...PartitionFallback</code>), correctness not asserted
P2	Config-update path	<code>STREAM.UPDATE</code> adding/removing sources, then assert only affected sources are recomputed and the rest preserved	<code>setStartingSequenceForSources</code> scoping
P2	Snapshot restore	restore a stream from snapshot/backup and assert resume	the cold-restore branch of recovery
P2	First-seq > 1	resume after age/limits expiry has advanced <code>FirstSeq</code>	off-by-one in the reverse-scan termination

Tier 0 (gated on implementation). When the persisted map lands, add a watermark matrix: `upToSeq == LastSeq` → trust; `upToSeq < LastSeq` → recompute (Tier 1); `upToSeq > LastSeq` → distrust; plus checksum-corruption and version-mismatch → fall through (degrade, not error); a crash between message durability and map flush → stale → Tier 1; clean `Stop` → Tier 0 hit; and a cluster snapshot round-trip.

Still to do

- The **P0/P1** items above — `...ResumeAfterLeaderStepDown` and the property/differential test are the highest-leverage next additions; the seeded store-state edges close the remaining hand-coverage gaps.

Resolved finding (was suspected pre-existing bug)

An earlier note suspected `setStartingSequenceForSources` mishandled transform sources because its Phase 2 sublist used `si.sfs` (transform **source** filters) rather than the **destination**. On inspection this was **not** a correctness bug: transform sources have an empty `FilterSubject`, so the old sublist inserted a `>` catch-all that matched the destination anyway (verified by `TestJetStreamStartingSequenceForSourcesTemplatedTransform`). It was purely an **efficiency** problem — the `>` defeated block skipping — now fixed by building the Phase 2 sublist from the destination wildcard form and dropping the dead `si.sfs` inserts.

11. Relationship to the durable-consumer proposal

This change is **orthogonal and complementary** to the durable-consumer/ `si.sseq` -persistence ideas in `sourcing-durable-resume.md`:

- Durable consumers / persisted `si.sseq` aim to **skip** resume work entirely (when state survives).
- This proposal makes the **fallback resume itself cheap** — which still runs on cold start, on snapshot restore, for limits/clustered streams, and any time persisted state is missing.

The two were measured head-to-head at 16 sources / 10M tail (see "At scale" above): a persisted resume map reads in ~2 μ s (pure in-memory, never touches the store), while the index recompute resolves in ~0.27 ms (one block load per source). Both are flat in store depth — versus ~1 s for the old reverse scan. The persisted map is the faster steady-state path, but it buys that speed with a write-path cost and a **new crash-consistency invariant**: the map must never lead durably-stored messages across truncation, compaction, message deletion, snapshot restore, and follower replication, or recovery resumes at a wrong sequence (gaps/duplicates). The index recompute carries none of that — the log plus the existing subject index remain the only source of truth — so it is the natural path to run whenever the persisted map is absent (pre-feature streams, cold start, restore) or must be revalidated.

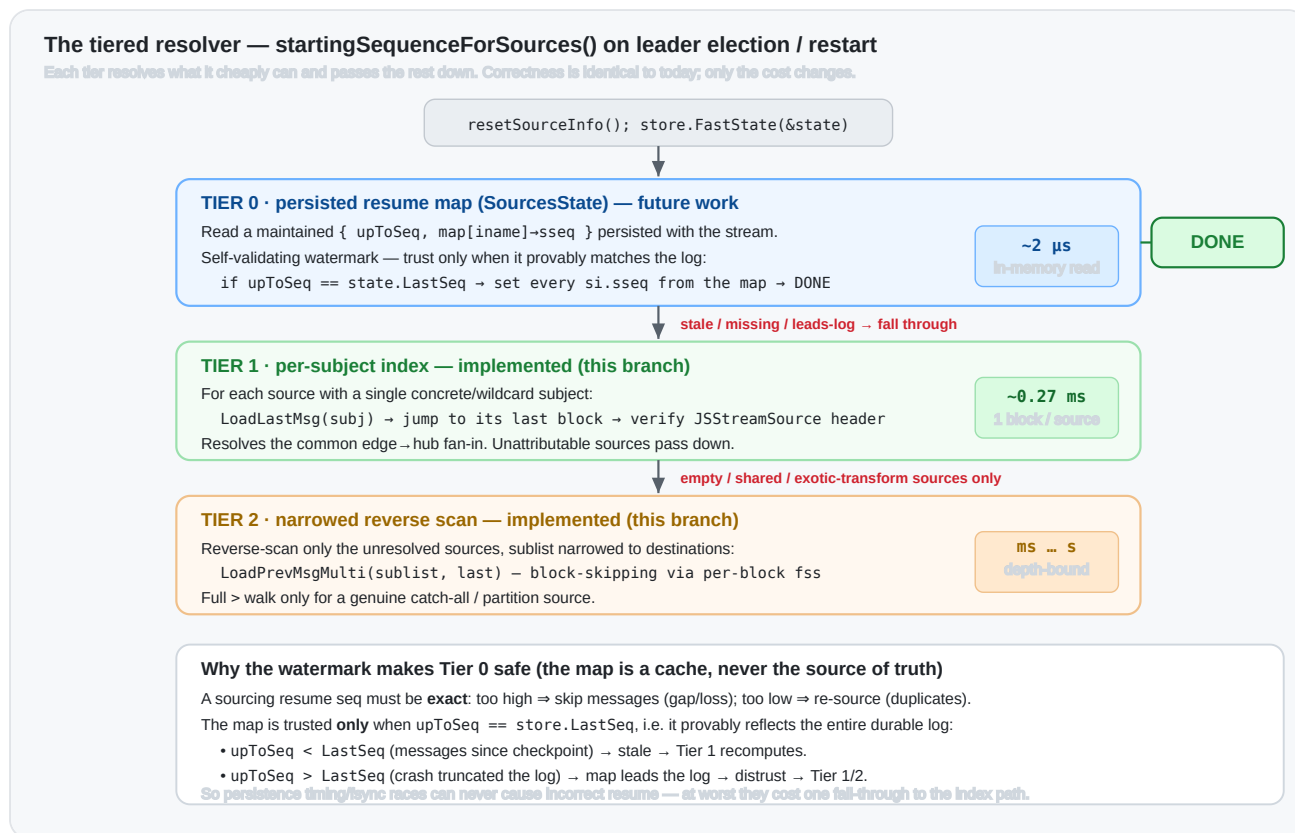
Recommended order: land this index-based resume first (self-contained, no protocol/state change, helps every retention type today, and replaces the $O(\text{depth})$ scan in the fallback the persisted design will still need), then layer the persistence/durable improvements on top.

12. Design sketch: Tier 0, a persisted resume map

This section sketches how the persisted-map approach (a parallel effort) would slot in as **Tier 0** on top of the resolver this branch already implements, what it buys, and the options/trade-offs for *where* the map lives. Nothing here is implemented on this branch — it is the design we'd build the two efforts toward.

12.1 The tiered resolver

`startingSequenceForSources` becomes three tiers, each resolving what it cheaply can and passing the remainder down. Correctness is identical at every tier — only the cost changes.



Tier	Mechanism	Resolves	Cost (16 src / 10M)	Status
0	read a persisted <code>{upToSeq, map[iname]→sseq}</code>	everything, if the map matches the log	~2 μs	design
1	<code>LoadLastMsg(subj)</code> per source via the subject index	distinct concrete/wildcard-subject sources	~0.27 ms	this branch
2	narrowed <code>LoadPrevMsgMulti</code> reverse scan	residual catch-all / exotic-transform sources	ms ... s	this branch

Tier 0 is a pure *fast path*: when present and trustworthy it returns immediately; otherwise the work falls through to the index recompute (Tier 1) and, for the genuine residue, the narrowed scan (Tier 2). Because Tiers 1–2 already exist and are cheap, Tier 0 can be added incrementally with no correctness risk to the fallback.

12.2 How Tier 0 works

The per-source resume sequence (`si.sseq` — the origin sequence of the last message sourced from each source) is already tracked in memory: it is set in `processInboundSourceMsg` (`stream.go`) as each sourced message is ingested, and today it is **always recomputed** from the store on leader election (`setLeader` → `setupSourceConsumers` → `startingSequenceForSources`). No source/mirror resume state is persisted anywhere today.

Tier 0 persists that in-memory map so recovery can *read* it instead of recomputing:

1. **Maintain** `map[iname]→sseq` — already done; it is the set of `si.sseq` values.
2. **Persist** the map, tagged with a watermark `upToSeq` = the hub stream `LastSeq` the map reflects.
3. **On recovery**, load the map and decide whether to trust it (§12.4); on a hit, set every `si.sseq` from the map and return — no store access at all.

12.3 The consistency requirement (why a naive map is unsafe)

A sourcing resume sequence must be **exact**, not merely "not ahead":

- **Map too high** (claims a seq the store doesn't durably have — e.g. a crash truncated messages written after the last checkpoint) ⇒ we resume *past* real messages ⇒ **gap / data loss**.
- **Map too low** (stale — messages were sourced since the last checkpoint) ⇒ we re-request from the origin and re-append ⇒ **duplicates** (sourced messages carry no `Nats-Msg-Id` , so the store's dedupe does not catch them).

So "persist it and trust it" is only safe if the map is *exactly* the store's state — which is precisely the crash-consistency burden a persisted map introduces, and why the state is recomputed today.

12.4 The watermark: self-validating trust

The cheap way to get exactness without per-source verification (which would just re-do Tier 1) is a single **watermark**. Persist `upToSeq` alongside the map and, on recovery, compare it to the store's actual `LastSeq` :

Condition	Meaning	Action
<code>upToSeq == LastSeq</code>	map provably reflects the entire durable log	trust it — Tier 0 hit (~µs)
<code>upToSeq < LastSeq</code>	messages were stored after the checkpoint	stale → fall to Tier 1
<code>upToSeq > LastSeq</code>	crash truncated the log below the checkpoint	map leads the log → distrust → Tier 1/2

This makes Tier 0 **safe by construction and independent of fsync timing**: if the last few sourced messages weren't durable and are lost on restart, the recovered `LastSeq` is below `upToSeq`, the map is distrusted, and Tier 1 recomputes. The worst case of *any* persistence race is one fall-through to the index path — never an incorrect resume. The common clean-shutdown / steady-checkpoint case (the typical restart and rolling-upgrade path) hits `upToSeq == LastSeq` and pays $\sim\mu\text{s}$.

12.5 Where to persist the map — options

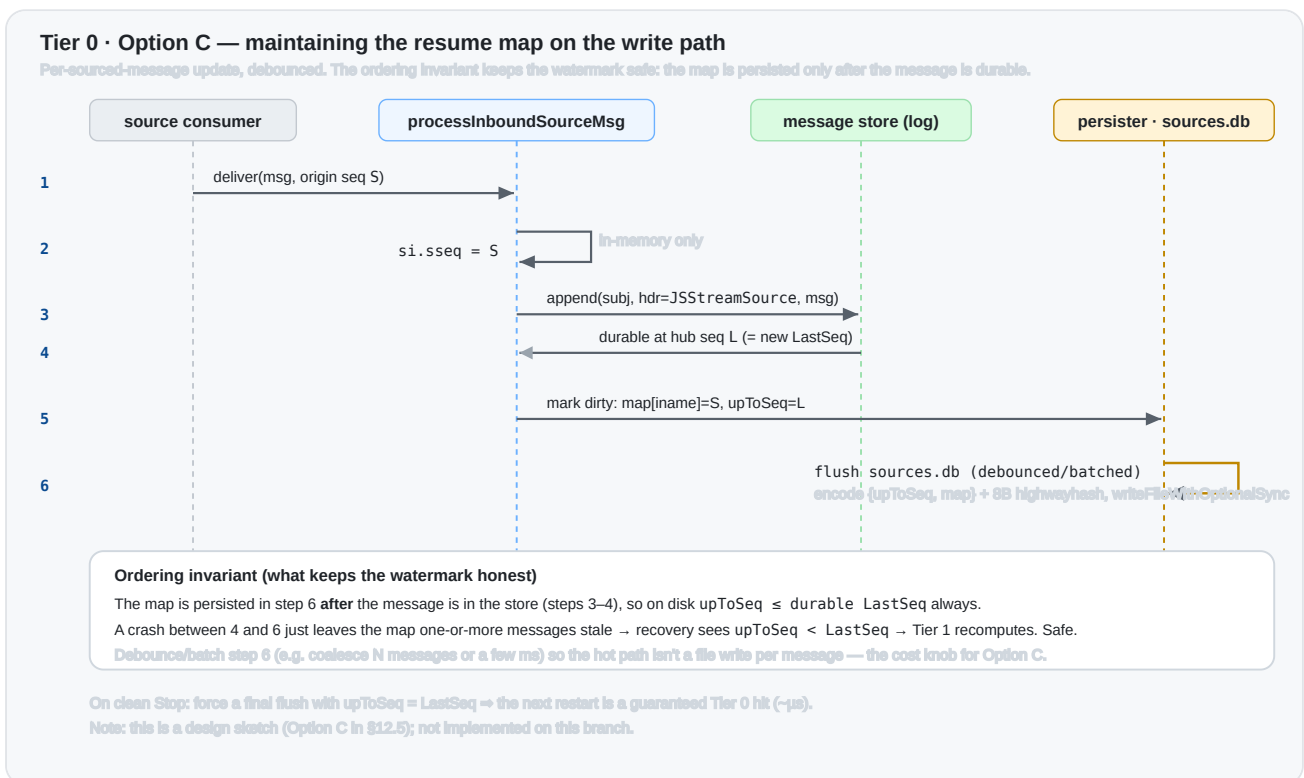
The map is tiny (`iname` $\rightarrow$ `uint64` , a few dozen entries), so the question is purely *where* it is written to stay consistent and *how often*. Five options, roughly increasing in consistency strength and in write-path / conflict cost:

Option	Mechanism	Pros	Cons
A. Inline in <code>index.db</code>	encode the map inside the filestore full-state file (<code>_writeFullState</code> / <code>recoverFullState</code>)	atomic with stream state; one file; watermark = the state's <code>LastSeq</code> for free	touches the checksummed filestore format (highest conflict surface); only as fresh as the periodic full-state write
B. Sidecar <code>sources.db</code>	a small separate file in the stream's <code>msgs/</code> dir, mirroring the consumer-state file pattern (<code>encodeConsumerState</code> / <code>writeState</code> / highwayhash sum)	self-contained; no change to <code>index.db</code> ; easy to version & checksum	a second file to write/sync and keep in step; its own staleness window
C. Write-path update	update + persist the map on every sourced message (the "always $\sim 2\ \mu\text{s}$ on recovery" design)	map is never stale (<code>upToSeq</code> $\approx$ <code>LastSeq</code> almost always) $\Rightarrow$ Tier 0 hit even after a crash	per-message write amplification; needs batching/debounce; strongest crash-consistency coupling
D. Periodic + on-stop	snapshot <code>{LastSeq, map}</code> on a timer and on clean <code>Stop</code>	trivial; no write-path cost; on-stop write makes clean restarts a guaranteed Tier 0 hit	after a <i>crash</i> , almost always stale $\Rightarrow$ falls to Tier 1 (still $\sim 0.27\ \text{ms}$, so acceptable)
E. RAFT snapshot (clustered)	add the map to the stream's replicated snapshot encoding	replicated to followers; survives leader	changes the replicated state format & version;

Option	Mechanism	Pros	Cons
	(StreamReplicatedState / stateSnapshot)	change without a recompute	must stay consistent across catchup/restore

Notes:

- **A vs B vs D differ only in the staleness window**, and the watermark makes any window *safe* — it only affects the Tier 0 *hit rate*, not correctness. So a low-risk first cut is **D** (periodic + on-stop, sidecar file), upgrading to **C** later if steady-state recovery latency on crash matters.
- **C** is what yields the headline “~2 μs even after a crash,” at the cost of touching the hot write path; it is the most invasive and the one most worth measuring for write-amplification. Its update sequence and the ordering invariant that keeps the watermark honest are shown below:



- **E** is orthogonal to A–D and is required for the *clustered* win — otherwise a clustered stream still recomputes (now via Tier 1) when a new leader has no local map. This is the largest piece and the one with the most format/version care.
- Whatever the choice, the map must be **bounded by LastSeq on read** (§12.4) and treated as a hint the log can always override.

12.6 Clustered streams

For replicated streams the resume state lives behind RAFT. Two sub-cases:

- **Same node re-elected / restarted** — a local sidecar/ `index.db` map (A/B/D) applies as in the single-server case.
- **Different node becomes leader** — it may have no local map (or a stale one). Either replicate the map in the stream snapshot (E), or simply let Tier 1 recompute on that node — which, post-this-branch, is ~0.27 ms rather than the old $O(\text{depth})$ scan. That fallback is what makes Tier 0 optional per-node rather than a hard dependency.

12.7 Expected performance

From the measured numbers (§10, "At scale"), with 16 sources over a 10M-message store:

Path	Recovery	Scales with	Flat in store depth?
Old reverse scan (pre-branch)	~0.97 s	store depth	no — $O(\text{depth})$
Tier 1 index recompute (this branch)	~0.27 ms	source count (~12 $\mu\text{s}/\text{src}$)	yes
Tier 0 persisted map (design)	~2 μs (reported)	source count (~0.13 $\mu\text{s}/\text{src}$)	yes

So Tier 0 is **~100× faster than Tier 1 in absolute terms** (a pure in-memory read vs one block load per source) and both are flat in depth. The practical question is whether ~0.27 ms recovery is already good enough: for most deployments it is, and Tier 1 alone removes the multi-second cliff. Tier 0 earns its keep when recovery happens *often* or with *many* sources — frequent leader elections / rolling restarts, or hundreds–thousands of sources where 0.27 ms → tens of ms (Tier 1) is worth driving back to μs . It does **not** change the asymptotics (both are $O(\text{sources})$); it lowers the constant by removing store I/O.

12.8 Recommendation

1. **Ship Tiers 1–2 now** (this branch): self-contained, no new state, removes the $O(\text{depth})$ cliff for every retention type, and is the fallback Tier 0 needs anyway.
2. **Add Tier 0 incrementally**: start with option **D** (periodic + on-stop sidecar, watermark-validated) for single-server/R1 — small, safe, and already turns clean restarts into μs . Measure the write path before considering **C**.
3. **Then Tier 0 for clusters** (option **E**): the largest and most consistency-sensitive piece; gate it behind the same watermark and keep Tier 1 as the per-node fallback so it can never resume incorrectly.

12.9 Concrete encoding (`SourcesState`)

The map is small, so a flat varint encoding mirroring `encodeConsumerState` (`store.go:401`) is plenty. It reuses the filestore conventions: a 2-byte header (`magic = 22` , `version = 1` , `hdrLen = 2` , `filestore.go:288–294`) and an 8-byte trailing highwayhash-64 checksum over the preceding bytes (`checksumSize = 8` , computed with the store's `fs.hh` digest as in `_writeFullState`).

Byte layout:

Offset	Field	Type	Notes
0	<code>magic</code>	<code>u8</code>	<code>= 22</code> (filestore magic)
1	<code>version</code>	<code>u8</code>	<code>= 1</code>
2...	<code>upToSeq</code>	<code>uvarint</code>	hub <code>LastSeq</code> the map reflects — the watermark (§12.4)
...	<code>n</code>	<code>uvarint</code>	number of source entries
...	<code>len(iname)</code>	<code>uvarint</code>	┐ repeated <code>n</code> times
...	<code>iname</code>	<code>bytes</code>	the source's unique index name
...	<code>sseq</code>	<code>uvarint</code>	┘ last origin sequence sourced for it
<code>end-8</code>	<code>checksum</code>	<code>[8]byte</code>	highwayhash-64 over <code>buf[0 : end-8]</code>

Encode (sketch):

```
const sourcesStateMagic = magic // 22, shared filestore magic
const sourcesStateVersion = uint8(1)

// encodeSourcesState serializes {upToSeq, map[iname]→sseq}. hh is the store's
// keyed highwayhash digest (fs.hh); pass nil to skip the checksum (tests).
func encodeSourcesState(upToSeq uint64, seqs map[string]uint64, hh hash.Hash64)
    // Upper bound: header + upToSeq + count + per-entry(len + iname + sseq)
    sz := hdrLen + 2*binary.MaxVarintLen64
    for iname := range seqs {
        sz += binary.MaxVarintLen64 + len(iname) + binary.MaxVarintLen64
    }
    sz += checksumSize
    buf := make([]byte, sz)

    buf[0], buf[1] = sourcesStateMagic, sourcesStateVersion
    n := hdrLen
    n += binary.PutUvarint(buf[n:], upToSeq)
```

```

    n += binary.PutUvarint(buf[n:], uint64(len(seqs)))
    for iname, sseq := range seqs {
        n += binary.PutUvarint(buf[n:], uint64(len(iname)))
        n += copy(buf[n:], iname)
        n += binary.PutUvarint(buf[n:], sseq)
    }
    if hh != nil {
        hh.Reset()
        hh.Write(buf[:n])
        n += copy(buf[n:], hh.Sum(nil)) // 8 bytes
    }
    return buf[:n]
}

```

Decode (sketch) — rejects a bad magic/version/checksum and returns the map plus the watermark, which `startingSequenceForSources` (Tier 0) then compares against `state.LastSeq`:

```

func decodeSourcesState(buf []byte, hh hash.Hash64) (upToSeq uint64, seqs map[string]uint64, err error) {
    if len(buf) < hdrLen+checksumSize || buf[0] != sourcesStateMagic || buf[hdrLen] != 0 {
        return 0, nil, errBadSourcesState
    }
    body, sum := buf[:len(buf)-checksumSize], buf[len(buf)-checksumSize:]
    if hh != nil {
        hh.Reset()
        hh.Write(body)
        if !bytes.Equal(hh.Sum(nil), sum) {
            return 0, nil, errBadSourcesState
        }
    }
    n := hdrLen
    upToSeq, c := binary.Uvarint(body[n:]); n += c
    cnt, c := binary.Uvarint(body[n:]); n += c
    seqs = make(map[string]uint64, cnt)
    for i := uint64(0); i < cnt; i++ {
        l, c := binary.Uvarint(body[n:]); n += c
        iname := string(body[n : n+int(l)]); n += int(l)
        sseq, c := binary.Uvarint(body[n:]); n += c
        seqs[iname] = sseq
    }
}

```

```
    return upToSeq, seqs, nil
}
```

Size: $\sim \text{len}(\text{iname}) + \sim 12$ bytes per source, so a 256-source stream is well under 10 KB – a single small write, well within one filesystem block for typical fan-ins.

Forward compatibility is the usual version bump (Tier 0 simply distrusts an unrecognized version and falls through to Tier 1, so an old/new format mismatch degrades to the index recompute rather than an error).

The Tier 0 read in `startingSequenceForSources` is then small:

```
// Tier 0: trust a persisted map only if it provably matches the durable log.
if buf, _ := mset.readSourcesState(); len(buf) > 0 {
    if upToSeq, seqs, err := decodeSourcesState(buf, mset.store.hh()); err == nil {
        for iname, sseq := range seqs {
            if si := mset.sources[iname]; si != nil {
                si.sseq, si.dseq = sseq, 0
            }
        }
        return // sources not in the map have never sourced → sseq stays 0
    }
}
// else fall through to Tier 1 (index recompute) / Tier 2 (narrowed scan)
```

Appendix — key references

Symbol	File:line	Role
<code>startingSequenceForSources</code>	<code>stream.go:4694</code>	the scan to replace (Phase 1 + Phase 2)
unconditional call	<code>stream.go:4821</code>	runs on every leader election / restart
<code>setStartingSequenceForSources</code>	<code>stream.go:4566</code>	update-path twin, same treatment
sublist build (stored subjects)	<code>stream.go:4742-4757</code>	defines a source's stored subject(s)
<code>LoadLastMsg</code> / <code>loadLast</code>	<code>filestore.go:9048</code> / <code>8939</code>	index-based last-msg-for-subject (returns header)

Symbol	File:line	Role
MultiLastSeqs	filestore.go:3806	batched last-seq-per-filter via index
psi / fs.psim	filestore.go:168 / 195	per-subject {total, fblk, lblk} ; store-wide subject → blocks index
mb.fss / SimpleState	filestore.go:239 / store.go:180	per-block subject → {Msgs, First, Last} index
psim maintenance on write	filestore.go:4933-4943	how total / lblk are kept current
LoadPrevMsgMulti	filestore.go:9403 / memstore.go:1991	the backward walk used by Phase 2
Nats-Stream-Source header	stream.go:635	constant; the per-message source provenance
genSourceHeader / streamAndSeq	stream.go:4470 / 4545	writes / parses origin stream, iname, origin seq

Tier 0 (design) persistence hooks:

Symbol	File:line	Role
processInboundSourceMsg (si.sseq = sseq)	stream.go:4371	where the in-memory resume map is updated per sourced message
setLeader → subscribeToStream → setupSourceConsumers	stream.go:1255 / 4957 / 4927	the recovery trigger Tier 0 would short-circuit
_writeFullState / recoverFullState	filestore.go:11681 / 1871	filestore full-state write/read (index.db) — option A host
encodeConsumerState / writeState	store.go:401 / filestore.go:13164	consumer-state file pattern to mirror for a sidecar sources.db (option B)
StreamReplicatedState / stateSnapshot	store.go:229 / jetstream_cluster.go:9897	replicated stream state / snapshot — option E

Symbol	File:line	Role
		host (no source state today)